Class #15 - Applications

1. Motivation: Machine learning uses linear models in situations where the machine is being *trained* to predict the outcome (_____variables) based on the values of the inputs (_____variables). The machine is given a set of training data where the values of the independent and dependent variables are _____. The machine then *learns* the _____between the independent variables and the dependent variables. One type of learing is to fit a curve, such as a least-squares line or parabola, to the data. Once the machine has learned the _____from the training data, it can then _____the value of the output based on a _____value for the input.

2. Example: Find the equation $y = a + bx$ of the least-squares line that best fits the data points

$$(2, 1), (5, 2), (7, 3), (8, 3).$$

3. Experimental data often produces points $(x_1, y_1), \cdots, (x_n, y_n)$ that, when graphed, seem to lie close to a line. We want to determine the numbers α and β so that the line $y = \alpha x + \beta$ is as close to the given points as possible. The *least-squares line* is the line that minimizes the sum of squares of the residuals.

4. Solve #2:

5. Least-Squares fitting of curves:
   Example: Suppose data points $(x_1, y_1), \cdots, (x_n, y_n)$ appear to lie along some sort of parabola instead of a straight line. Suppose we wish to approximate the data by an equation of the form

$$y = a + bx + cx^2.$$

   Describe the linear model (a system of linear equations) that produces a "least-squares fit" of the data points.